

Haris Zubair

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Technical Skills

Hardware: Nexys 3/A7 FPGA, NVIDIA Jetson Xavier, Arduino, Raspberry Pi 4, ESP32, STM32, FANUC, KUKA

Programming: Python, C++, C, React Native, MATLAB, Java, SQL, VHDL, Verilog, ARM Assembly

Tools: Xilinx Vivado Design Suite, Cadence Virtuoso, AutoCAD, SolidWorks (CSWA Certificate), KiCad, LTspice, Bambu Studio, Git, Ubuntu (Linux), ROS2, Visual Studio Code, GitHub, Microsoft Office Products, WSL2, Docker, STM32CubeIDE, Azure, AWS

Education

Bachelor of Engineering, Computer Engineering (Co-op)

September 2023 – May 2028

University of Guelph: Department of Computer and Electrical Engineering, Guelph ON

- On the College of Engineering Dean's Honour List (3.7 GPA)
- **Courses:** Engineering Design II, Computer Organization, Digital System Design, Electric Circuits, Microcomputer Interfacing, Electronic Devices, Signal Processing, Data Structures, Micro and Nano-Scale Electronics

Experience

AI Software Engineering Student (Electrical Design Software)

January 2026 – September 2026

Hammond Power Solutions, Guelph ON

- Developed AI tools with Python to accelerate the design of electrical transformers
- Trained an FT-Transformer (PyTorch) and FAISS model for product similarity matching on transformer attributes
- Used Azure OpenAI and AI Foundry to build AI agents that retrieve, interpret, and summarize engineering data
- Received training on understanding electrical transformer theory, design, and practical usage

IoT Developer: Biomedical Robotics Applications

September 2025 – Current

Robotics Institute: Rehabilitation and Assistive Robotics Devices, Guelph ON

- Worked on a research project which involves creating an app for wireless control (BLE) of a hardware system
- Tested a React Native application on Android and iOS devices, using Xcode to deploy builds and verify functionality
- Researching and prototyping PCB designs for an IoT board based on an ESP32 microcontroller for relay control
- Conducted electrical testing in a research lab setting to verify electronic assemblies after soldering
- Troubleshoot AC-powered automation systems while following electrical safety protocols and reviewing datasheets

Vice President Safety, Electrical Team Member (Club)

September 2023 – Current

University of Guelph Robotics Team, Guelph ON

- Designed various electrical systems (E-Stop, wiring harness, sensors) and programmed microcontrollers for a rover
- Communicated with 4 sub-teams to solve logistical and design issues, as well as implementing solutions
- Created a safety report for the rover's electrical system and safety features, created electrical schematics in AutoCAD
- Used various electrical tools (multimeter, oscilloscope, soldering iron) to complete and troubleshoot projects

URA: Machine Learning for Medical Image Analysis Research Position

May 2025 – August 2025

Dr. Eran Ukwatta's AI-Enabled Medical Image Analysis Lab, Guelph ON

- Worked on an intracranial hemorrhage detection deep learning model that utilized 42,000 DICOM images
- Developed Python TensorFlow Keras model frameworks for ensemble model, utilized Excel for data organization
- Used Pandas, Matplotlib, and NumPy to reformat data frames and create graphs for data visualization
- Learned in detail about math/statistics behind ML/DL concepts and proper model architecture

IEEE Student Branch Membership Development Coordinator and Hardware Lead (Club)

February 2025 – Current

IEEE Student Branch, Guelph ON

- Planned multiple technical events on behalf of IEEE and taught other students about electronics using breadboards
- Designed car chassis and camera gimbal in SolidWorks and set up electronics/software for interactive workshops

Engineering Projects

Full CPU with VHDL/FPGA Board, and Hardware Datapath

November 2024 – December 2024

- Designed a CPU and implemented it in VHDL using a hierarchical structure, CPU takes in 16-bit instructions
- Implemented CPU onto FPGA board, used Xilinx Vivado tools to complete all tasks, learned about CPU architecture
- Developed VHDL test benches (waveform) and used simulation to validate instruction execution/control signal timing
- Designed and assembled a Datapath that contained an ALU, registers, and storage, switches used as control signals

Wi-Fi Controlled ESP32 RC Car

April 2025 – May 2025

- Designed a small RC car using an ESP32-WROVER board, 4 micro motors, and a TB6612FNG motor drive board
- Programmed in C++ to be controllable via Wi-Fi WebSocket using a web application (JS and HTML for UI)
- 3D modelled and printed chassis to hold all components in SolidWorks

Reverse Engineered SolidWorks Assembly Model

January 2025 – April 2025

- Led a 4-person team to reverse engineer and digitally replicate a solar-powered ostrich model
- Measured each component using calipers and precision tools to collect accurate dimensions for 3D modeling
- Designed parts in SolidWorks and produced detailed 2D technical drawings based on collected dimensions
- Built a functional digital assembly with 69 unique parts and 138 total parts, developed an animated exploded view

STM32G0B1CET6 CAN-FD Arm-Motor Drive Board PCB

October 2025

- Designed and soldered a 4-layer PCB to control Dynamixel and ODrive motors via CAN-FD using an STM32G0B1CET6
- Set up CAN transceiver modules, utilized a MAX485 IC for UART to RS485 communication with Dynamixel motors
- Designed a 12V to 5V buck converter, and a 5V to 3.3V LDO circuit to deliver appropriate voltage to each device
- Learned how to use CAN bus protocol and how to set up electronics for CAN bus usage

STM32 NUCLEO-F767ZI Supported Multi-Motor Drive Board PCB

March 2025

- Designed and soldered a 4-layer PCB to connect motion controllers to an STM32 NUCLEO board for motor control
- Board was designed to communicate over UART and I2C, connected STM pins to JST-XH connectors
- Board was designed in KiCad to be through-hole soldered, organized all connections to fit onto a small PCB

Automated 3-Axis Robot Arm

December 2023 – August 2025

- Designed a programmable arm controlled by servos and potentiometers using C++ and an Arduino Uno Rev3
- Implemented control logic so arm could complete repeated tasks automatically or controlled with live input
- Updated to use ROS2 for controls with Topics-based communication and have motions planned in Gazebo simulation

Team Object-Oriented Programming Project

January 2024 – April 2024

- Worked with a 5-person team on a game programming project that was implemented in 3 phases using GitHub
- Project followed object-oriented programming principles in Java, and created GUI from scratch
- Made use of Java Swing and Java AWT for UI components, learned how to create UIs with detail

Certifications

DeepLearning.AI Agentic AI Course Certificate

January 2026

- Completed Agentic AI Learning course, learned about agentic design patterns using Python

Kaggle Course Certificates

March 2025

- Pandas, Introduction to Python, Introduction to Machine Learning, Intermediate Machine Learning, Deep Learning, Computer Vision

SolidWorks CSWA Certificate

March 2025

- Obtained via completion and passing of the SolidWorks CSWA Exam

IEEE Circuits and Systems Society CASSBLITZ Certificate of Completion

May 2025

- Obtained via completion and passing of the IEEE CASSBLITZ week digital/analog circuits exam

Lab and Workplace Safety Courses

May 2025

- Completed EHS Worker Health and Safety Refresher Training, WHMIS 2025, and EHS Laboratory Safety courses